

EXPERIMENT - 29

1. Problem Statement

Write a C program to perform the following menu-driven operations:

- Count even numbers, odd numbers, positive, negative and zero elements in an array.
- Search for an element in an array using linear search.
- Merge 2 arrays.
- Exit.

2. Algorithm

Main Program

Step 1: START

Step 2: Display the menu and read choice.

Step 3: Using a switch statement on choice:

 If choice is 1, call countElements().

 If choice is 2, call linearSearch().

 If choice is 3, call mergeArrays().

 If choice is 4, display "Exiting the program."

Step 4: Repeat from Step 2 while choice is not equal to 4.

Step 5: STOP

Function countElements()

Step 1: Read n and then read n elements into the array a.

Step 2: Set even, odd, positive, negative and zero all to 0.

Step 3: Repeat for $i = 0$ to $n - 1$:

 If $a[i]$ divided by 2 leaves remainder 0, increase even by 1, otherwise increase odd by 1.

 If $a[i]$ is greater than 0 increase positive by 1, else if $a[i]$ is less than 0 increase negative by 1, otherwise increase zero by 1.

Step 4: Display all five counts.

Step 5: Return to the calling function.

Function linearSearch()

Step 1: Read n and then read n elements into the array a.

Step 2: Read the element to be searched, key, and set found = 0.

Step 3: Repeat for $i = 0$ to $n - 1$:

 If $a[i]$ is equal to key, display the position $i + 1$ and set found = 1.

Step 4: If found is still 0, display that the element is not present.

Step 5: Return to the calling function.

Function mergeArrays()

Step 1: Read $n1$ and then read $n1$ elements into the array a .

Step 2: Read $n2$ and then read $n2$ elements into the array b .

Step 3: Set $k = 0$.

Step 4: Repeat for $i = 0$ to $n1 - 1$: set $c[k] = a[i]$ and increase k by 1.

Step 5: Repeat for $i = 0$ to $n2 - 1$: set $c[k] = b[i]$ and increase k by 1.

Step 6: Display the array c from $c[0]$ to $c[k - 1]$.

Step 7: Return to the calling function.

3. Test Case Table

#	Operation	Input	Expected Output	Actual Output	Result
1	1. Count	12, -5, 0, 7, -8, 3, 0	Even 4, Odd 3, Positive 3, Negative 2, Zero 2	Same	PASS
2	1. Count	-3, -7, -11	Odd 3 and Negative 3, the rest 0	Same	PASS
3	2. Search	10, 25, 7, 40, 7 key = 7	Found at position 3 and again at position 5	Both positions	PASS
4	2. Search	10, 25, 7, 40, 7 key = 99	99 is not present in the array	Same	PASS
5	3. Merge	{1, 2, 3} and {4, 5}	1 2 3 4 5	1 2 3 4 5	PASS
6	3. Merge	{5} and {9}	5 9	5 9	PASS
7	4. Exit	choice = 4	Exiting the program, and the loop ends	Program ended	PASS

4. Source Code

arrayops.c

```
/* Experiment 29. Menu driven program that works on arrays. The switch
   statement picks the operation and the menu keeps coming back until
   the user chooses Exit. */

#include <stdio.h>
#define MAX 100

void countElements(void);
void linearSearch(void);
void mergeArrays(void);

int main(void)
{
    int choice;

    do {
        printf("\nMENU\n");
        printf("1. Count even, odd, positive, negative and zero elements\n");
        printf("2. Search for an element using linear search\n");
        printf("3. Merge 2 arrays\n");
        printf("4. Exit\n");
        printf("Enter your choice: ");
        scanf("%d", &choice);

        switch (choice) {
            case 1: countElements(); break;
            case 2: linearSearch(); break;
            case 3: mergeArrays(); break;
            case 4: printf("Exiting the program.\n"); break;
        }
    } while (choice != 4);

    return 0;
}

/* Every element is checked twice, once for even or odd and once for
   its sign. Zero is counted as even, and it is neither positive nor
   negative, so it gets its own counter. */
void countElements(void)
{
    int a[MAX], n, i;
    int even = 0, odd = 0, positive = 0, negative = 0, zero = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    for (i = 0; i < n; i++) {
        if (a[i] % 2 == 0)
            even++;
        else
            odd++;

        if (a[i] > 0)
            positive++;
        else if (a[i] < 0)
            negative++;
        else
            zero++;
    }

    printf("Even elements      = %d\n", even);
    printf("Odd elements       = %d\n", odd);
    printf("Positive elements = %d\n", positive);
}
```

```

    printf("Negative elements = %d\n", negative);
    printf("Zero elements      = %d\n", zero);
}

/* The array is checked one place at a time from the beginning. A flag
   remembers whether the element was found, so the not-found message is
   printed only after the whole array has been looked through. */
void linearSearch(void)
{
    int a[MAX], n, i, key, found = 0;

    printf("Enter the number of elements: ");
    scanf("%d", &n);

    printf("Enter %d elements: ", n);
    for (i = 0; i < n; i++)
        scanf("%d", &a[i]);

    printf("Enter the element to be searched: ");
    scanf("%d", &key);

    for (i = 0; i < n; i++) {
        if (a[i] == key) {
            printf("%d found at position %d\n", key, i + 1);
            found = 1;
        }
    }

    if (found == 0)
        printf("%d is not present in the array\n", key);
}

/* The first array is copied into the third one and then the second
   array is put after it, so k keeps counting forward across both
   copies. */
void mergeArrays(void)
{
    int a[MAX], b[MAX], c[2 * MAX];
    int n1, n2, i, k;

    printf("Enter the number of elements in the first array: ");
    scanf("%d", &n1);
    printf("Enter %d elements: ", n1);
    for (i = 0; i < n1; i++)
        scanf("%d", &a[i]);

    printf("Enter the number of elements in the second array: ");
    scanf("%d", &n2);
    printf("Enter %d elements: ", n2);
    for (i = 0; i < n2; i++)
        scanf("%d", &b[i]);

    k = 0;
    for (i = 0; i < n1; i++) {
        c[k] = a[i];
        k++;
    }
    for (i = 0; i < n2; i++) {
        c[k] = b[i];
        k++;
    }

    printf("Merged array: ");
    for (i = 0; i < k; i++)
        printf("%d ", c[i]);
    printf("\n");
}

```

5. Compilation and Output

I compiled it with `gcc -Wall -Wextra arrayops.c -o arrayops` so that even the small warnings would show up, and it compiled clean in the first attempt itself. The screenshots below are of the runs I did on my own machine.

Counting the Elements

```
adilshakil@ADILs-MacBook-Air MyAssignment % gcc -Wall -Wextra arrayops.c -o arrayops
adilshakil@ADILs-MacBook-Air MyAssignment % ./arrayops

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 1
Enter the number of elements: 7
Enter 7 elements: 12 -5 0 7 -8 3 0
Even elements    = 4
Odd elements     = 3
Positive elements = 3
Negative elements = 2
Zero elements    = 2

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 1
Enter the number of elements: 3
Enter 3 elements: -3 -7 -11
Even elements    = 0
Odd elements     = 3
Positive elements = 0
Negative elements = 3
Zero elements    = 0
```

Linear Search, Found and Not Found

```
adilshakil@ADILs-MacBook-Air MyAssignment % ./arrayops

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 2
Enter the number of elements: 5
Enter 5 elements: 10 25 7 40 7
Enter the element to be searched: 7
7 found at position 3
7 found at position 5

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 2
Enter the number of elements: 5
Enter 5 elements: 10 25 7 40 7
Enter the element to be searched: 99
99 is not present in the array
```

Merging Two Arrays

```
adilshakil — -zsh — 120x30

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 3
Enter the number of elements in the first array: 3
Enter 3 elements: 1 2 3
Enter the number of elements in the second array: 2
Enter 2 elements: 4 5
Merged array: 1 2 3 4 5

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 3
Enter the number of elements in the first array: 1
Enter 1 elements: 5
Enter the number of elements in the second array: 1
Enter 1 elements: 9
Merged array: 5 9
█
```

Exit

```
adilshakil — -zsh — 120x30

MENU
1. Count even, odd, positive, negative and zero elements
2. Search for an element using linear search
3. Merge 2 arrays
4. Exit
Enter your choice: 4
Exiting the program.
adilshakil@ADILs-MacBook-Air MyAssignment %
█
```

All seven test cases came out right. While testing the counting I checked that zero is put in the even group and also in its own group, because it is neither positive nor negative, so the two sets of counts each add up to the number of elements. I also wrote the even test as the remainder being 0 rather than the odd test being 1, because in C a negative odd number like -5 gives -1 as the remainder and would have been counted wrongly.